For $10 \leq E_\gamma \leq 40$ MeV the photon wavelength is comparable to the nuclear size. It is therefore easily absorbed, which causes collective nuclear vibrational motions (so-called dipole vibrations, because the neutrons and protons are assumed to vibrate in separate groups). De-excitation occurs through $\gamma$-emission. This is known as the "giant resonance" region, because the total cross-section for heavier nuclides goes up to hundreds of millibarns. For higher $E_\gamma$, nucleons may be expelled, the main reactions being ($\gamma$,n), ($\gamma$,2n) and ($\gamma$,np) in order of descending importance. As the $\gamma$-energy increases and the wavelength decreases to nucleon dimensions, interaction with nuclear groups (e.g. deuterons) or single nucleons takes place. Below 550 MeV one pion plus a nucleon may be emitted in the de-excitation following the photon absorption. At higher energies, several pions may be emitted. Very little is known about the details of these processes.

## 11.9. EXERCISES

**11.1.** A 0.01 mm thick gold foil, 1 $cm^2$ in area, is irradiated with thermal neutrons. The (n,$\gamma$) cross-section is 99 b. What is the transformation rate at a n-flux of $10^{19}$ n $m^{-2}$ $s^{-1}$?

**11.2.** Assume the irradiation time of the gold foil in the previous problem is one week. What percentage of the original gold atoms in the target have undergone transformation?

**11.3.** A water-cooled copper foil (0.1 mm thick) is irradiated by the internal beam of a sector focused cyclotron with 1.2 mA $H^+$ ions of 24 MeV for 90 min. The reaction $^{63}$Cu(p,pn)$^{62}$Cu occurs with a probability of 0.086 b. Copper consists to 69% of $^{63}$Cu. The proton beam has a cross-section of only 15 $mm^2$. (a) How many $^{62}$Cu atoms have been formed? (b) What fraction of the projectiles have reacted to form $^{62}$Cu? (c) What cooling effect is required (kW) at the target?

**11.4.** Calculate the macroscopic cross-section for reaction of natural uranium with thermal neutrons. See Figures 14.1 and 19.5.

**11.5.** (a) Estimate the yield (% of fissions) of $^{142}$La in thermal fission of $^{235}$U given a chain yield of 5.839% for A = 142, most probable charge = 55.86 and a width parameter of 0.56. (b) Is an appreciable amount of $^{142}$Nd formed directly in thermal fission of $^{235}$U?

**11.6.** The total kinetic energy of the fragments from thermal fission of $^{239}$Pu is 177.7 MeV and the average fragment masses are 100.34 and 139.66 u respectively. What are the kinetic energies of the average light and heavy mass fragments?

**11.7.** Calculate the kinetic energy of the $^4$He ion formed through thermal neutron capture in $^{10}$B.

**11.8.** What is the minimum photon energy required for the reaction $^{11}$B($\gamma$,n)$^{10}$B?

**11.9.** A 2 MeV neutron collides elastically with an iron atom ($^{56}$Fe). What is the average temperature (corresponding to the maximum velocity) which can be ascribed to the iron nucleus after the collision?

## 11.10. LITERATURE

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